

MAT31-1

Linear Algebra III

Euclidean spaces

The scalar product, the geometry it enables, and the spectral theorem.

Albert School — 30 hours

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1 The scalar product

Throughout, E is a real vector space (MAT11-3).

Definition 1 (Scalar product) A **scalar product** (or **inner product**) on E is a map $\langle \cdot, \cdot \rangle : E \times E \rightarrow \mathbb{R}$ satisfying, for all $u, v, w \in E$ and all $\lambda \in \mathbb{R}$:

- **Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$.
- **Bilinearity:** it is linear in each argument; by symmetry it suffices that

$$\langle \lambda u + w, v \rangle = \lambda \langle u, v \rangle + \langle w, v \rangle.$$

- **Positive definiteness:** $\langle u, u \rangle \geq 0$ for all u , with $\langle u, u \rangle = 0$ if and only if $u = 0$.

A real vector space equipped with a scalar product is a **Euclidean space** (when finite-dimensional).

1.1 Examples

- On $\mathbb{R}^n$, the **standard scalar product**

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i = x^T y.$$

- On the space $\mathbb{R}[X]_{\leq n}$ of polynomials of degree at most n , fixing distinct reals $a_0, \dots, a_n$,

$$\langle P, Q \rangle = \sum_{i=0}^n P(a_i) Q(a_i),$$

or, using a real interval $[a, b]$,

$$\langle P, Q \rangle = \int_a^b P(t) Q(t) dt.$$

- On the matrix space $M_n(\mathbb{R})$, the **Frobenius scalar product**

$$\langle A, B \rangle = \text{tr}(A^T B) = \sum_{i,j} a_{ij} b_{ij}.$$

2 Norm and the Cauchy–Schwarz inequality

Definition 2 The **norm** attached to a scalar product is

$$\|u\| = \sqrt{\langle u, u \rangle}.$$

Theorem 3 (Cauchy–Schwarz) For all $u, v \in E$,

$$|\langle u, v \rangle| \leq \|u\| \|v\|,$$

with equality if and only if u and v are collinear.

Proposition 4 (Norm properties) For all $u, v \in E$ and $\lambda \in \mathbb{R}$:

- $\|u\| \geq 0$, with $\|u\| = 0$ iff $u = 0$;
- $\|\lambda u\| = |\lambda| \|u\|$;
- **Triangle inequality**: $\|u + v\| \leq \|u\| + \|v\|$.

2.1 Geometric interpretation

By Cauchy–Schwarz, for non-zero u, v the quotient $\langle u, v \rangle / (\|u\| \|v\|)$ lies in $[-1, 1]$. The **angle** $\theta \in [0, \pi]$ between u and v is defined by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

This quantity is the **cosine similarity** of u and v .

3 Orthogonality

Definition 5 Two vectors $u, v \in E$ are **orthogonal**, written $u \perp v$, if $\langle u, v \rangle = 0$.

A family $(u_1, \dots, u_p)$ is:

- **orthogonal** if $\langle u_i, u_j \rangle = 0$ for all $i \neq j$;
- **orthonormal** if it is orthogonal and $\|u_i\| = 1$ for all i , i.e. $\langle u_i, u_j \rangle = \delta_{ij}$.

Proposition 6 (Automatic linear independence) Every orthogonal family of **non-zero** vectors is linearly independent. In particular every orthonormal family is linearly independent.

Proposition 7 (Pythagorean theorem) If $u \perp v$ then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

If $(e_1, \dots, e_n)$ is an orthonormal basis of E , then any $u \in E$ has coordinates given by scalar products:

$$u = \sum_{i=1}^n \langle u, e_i \rangle e_i, \quad \|u\|^2 = \sum_{i=1}^n \langle u, e_i \rangle^2.$$

4 The Gram–Schmidt process

Theorem 8 (Gram–Schmidt) Let $(v_1, \dots, v_p)$ be a linearly independent family of E . Define

$$w_1 = v_1, \\ w_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, w_j \rangle}{\langle w_j, w_j \rangle} w_j \quad (k = 2, \dots, p).$$

Then $(w_1, \dots, w_p)$ is an orthogonal family, and normalising $e_k = w_k / \|w_k\|$ gives an orthonormal family. For every k ,

$$\text{span}(e_1, \dots, e_k) = \text{span}(v_1, \dots, v_k),$$

so the process preserves successive spans.

Applied to a basis, Gram–Schmidt produces an **orthonormal basis**. The procedure applies verbatim in polynomial and function spaces equipped with one of the scalar products above.

5 Orthogonal matrices

Definition 9 A matrix $P \in M_n(\mathbb{R})$ is **orthogonal** if

$$P^T P = P P^T = I, \quad \text{i.e.} \quad P^T = P^{-1}.$$

Proposition 10 P is orthogonal if and only if its columns form an orthonormal basis of $\mathbb{R}^n$ for the standard scalar product. Orthogonal matrices preserve the scalar product and the norm: $\langle Px, Py \rangle = \langle x, y \rangle$ and $\|Px\| = \|x\|$. The product of two orthogonal matrices is orthogonal, and so is the inverse.

Proposition 11 (Rotations vs. reflections) If P is orthogonal then $\det P = \pm 1$. One calls P a **rotation** (direct isometry) when $\det P = +1$ and a **reflection** (indirect isometry) when $\det P = -1$.

6 Symmetric endomorphisms

Definition 12 A matrix $A \in M_n(\mathbb{R})$ is **symmetric** if $A^T = A$. The associated endomorphism $x \mapsto Ax$ is then called a **symmetric endomorphism**; it satisfies

$$\langle Ax, y \rangle = \langle x, Ay \rangle \quad \text{for all } x, y.$$

Proposition 13 Let A be a real symmetric matrix.

- Every eigenvalue of A is **real**.
- Eigenvectors associated with **distinct** eigenvalues are **orthogonal**; that is, the eigenspaces E_λ and E_μ are orthogonal for $\lambda \neq \mu$.

7 The spectral theorem

Theorem 14 (Spectral theorem) Every real symmetric matrix $A \in M_n(\mathbb{R})$ is **orthogonally diagonalizable**: there exist an orthogonal matrix P ($P^T = P^{-1}$) and a diagonal matrix D with

$$A = PDP^T.$$

The diagonal entries of D are the (real) eigenvalues of A , and the columns of P form an orthonormal basis of eigenvectors.

Geometrically, the transformation $x \mapsto Ax$ acts by independent scalings, by the eigenvalues, along the mutually orthogonal directions given by the columns of P .

7.1 Procedure

To orthogonally diagonalize a symmetric A :

1. compute the eigenvalues from χ_A and a basis of each eigenspace (MAT21-1);
2. apply Gram–Schmidt within each eigenspace to obtain an orthonormal basis of it (eigenspaces for distinct eigenvalues are already orthogonal);
3. collect these vectors as the columns of P and place the eigenvalues on the diagonal of D ; then $A = PDP^T$.

8 Positive definiteness

Definition 15 A symmetric matrix $A \in M_n(\mathbb{R})$, with associated **quadratic form** $q(x) = x^T Ax$, is classified by the signs of its eigenvalues $\lambda_1, \dots, \lambda_n$ (all real, by the spectral theorem):

- **positive definite** if all $\lambda_i > 0$ (equivalently $q(x) > 0$ for all $x \neq 0$);
- **positive semi-definite** if all $\lambda_i \geq 0$;
- **negative definite / semi-definite** if all $\lambda_i < 0$ / $\lambda_i \leq 0$;
- **indefinite** if it has both a positive and a negative eigenvalue.

This classification applied to the **Hessian matrix** of a twice-differentiable function determines the nature of a critical point: a positive definite Hessian gives a local minimum, a negative definite Hessian a local maximum, and an indefinite Hessian a saddle point.

9 Orthogonal complement

Definition 16 Let F be a subspace of E . Its **orthogonal complement** is

$$F^\perp = \{u \in E : \langle u, v \rangle = 0 \text{ for all } v \in F\}.$$

It is a subspace of E .

To compute $F^\perp$: if $F = \text{span}(v_1, \dots, v_p)$, then $u \in F^\perp$ iff $\langle u, v_i \rangle = 0$ for every i , a homogeneous linear system.

Theorem 17 (Orthogonal decomposition) If E is finite-dimensional and F is a subspace, then

$$E = F \oplus F^\perp, \quad \dim F^\perp = \dim E - \dim F,$$

$$\text{and } (F^\perp)^\perp = F.$$

Thus every $u \in E$ decomposes uniquely as $u = u_F + u_{F^\perp}$ with $u_F \in F$ and $u_{F^\perp} \in F^\perp$.

10 Orthogonal projection

Definition 18 In the decomposition $E = F \oplus F^\perp$, the **orthogonal projection** onto F is the map $p_F : E \rightarrow F$ sending $u = u_F + u_{F^\perp}$ to u_F .

Proposition 19 (Projection via an orthonormal basis) If $(e_1, \dots, e_k)$ is an orthonormal basis of F , then for every $u \in E$,

$$p_F(u) = \sum_{i=1}^k \langle u, e_i \rangle e_i.$$

In $\mathbb{R}^n$, writing $Q = (e_1 \mid \dots \mid e_k)$, the **projection matrix** onto F is

$$P_F = QQ^T,$$

so that $p_F(u) = P_F u$. It satisfies $P_F^T = P_F$ and $P_F^2 = P_F$.

Theorem 20 (Best approximation) For every $u \in E$, the projection $p_F(u)$ is the unique element of F closest to u :

$$\|u - p_F(u)\| \leq \|u - v\| \quad \text{for all } v \in F,$$

with equality only for $v = p_F(u)$.

Proof. For any $v \in F$, write $u - v = (u - p_F(u)) + (p_F(u) - v)$. The first term lies in $F^\perp$ and the second in F , so they are orthogonal; by the Pythagorean theorem,

$$\|u - v\|^2 = \|u - p_F(u)\|^2 + \|p_F(u) - v\|^2 \geq \|u - p_F(u)\|^2,$$

with equality iff $p_F(u) - v = 0$. □

11 Distance to a subspace

Definition 21 The **distance** from $u \in E$ to a subspace F is

$$d(u, F) = \min_{v \in F} \|u - v\| = \|u - p_F(u)\| = \|u_{F^\perp}\|.$$

By the best-approximation theorem this minimum is attained at the orthogonal projection. This is the geometric idea behind **least-squares regression**: fitting a model by minimizing the distance from the data vector to the subspace of achievable predictions amounts to projecting the data vector orthogonally onto that subspace.